Name

Mathematics Advanced Year 12

Differentiating Exponential Functions

Book 1 Differentiation with exponential functions

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-12-04 selects and applies differentiation methods to solve problems

Differentiation with exponential functions

- Apply the rules of differentiation to find the derivative of $f(x) = ke^{ax}$, where k and a are constants
- Use the chain rule to prove $\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$, where a and b are constants and $a \neq 0$
- Prove and use the formula $\frac{d}{dx}(a^x) = (\ln a)a^x$, where a is a constant and $a > 0$
- Use the chain rule to differentiate functions of the form $e^{f(x)}$

Derivative of Exponential Functions

Derivative of Exponential Functions

The derivative of a natural exponential function is itself.

$$f(x) = e^x$$
$$f'(x) = e^x$$

Identify the derivative of an exponential function,

$f(x) = e^x$	$f(x) = 3e^x$	$f(x) = 4e^x - 10x$
$f'(x) = e^x$	$f'(x) = 3e^x$	$f'(x) = 4e^x - 10$

Differentiate these functions.

a	$f(x) = 5e^x$	**b**	$f(x) = -18e^x$	**c**	$f(x) = e^x + 24x^2$
d	$f(x) = e^5$	**e**	$f(x) = 17e^x + ex$	**f**	$f(x) = ke^x$

Investigation Apply chain rule to differentiate $f(x) = e^{ax+b}$

Using chain rule $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$, differentiate $f(x) = e^{3x+4}$.

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$=$

$=$

$=$

Let $u = 3x + 4$

$\frac{du}{dx} =$

Let $y = e^u$

$\frac{dy}{du} =$

Using chain rule $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$, differentiate $f(x) = e^{-7x+11}$.

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$=$

$=$

$=$

Let $u =$

$\frac{du}{dx} =$

Let $y =$

$\frac{dy}{du} =$

What can you conclude about the derivative of $f(x) = e^{ax+b}$? $f'(x) =$ .....

Derivative of Exponential Functions

$$f(x) = e^{ax+b}$$

$$f'(x) = ae^{ax+b}$$

Identify the derivative of an exponential function,

$f(x) = e^{9x}$	$f(x) = e^{x+9}$	$f(x) = e^{-5x+7}$	$f(x) = 3e^{5x-11}$
$f'(x) = 9e^{9x}$	$f'(x) = e^{x+9}$	$f'(x) = -5e^{-5x+7}$	$f'(x) = 15e^{5x-11}$

Differentiate these functions.

a $f(x) = e^{5x}$	**b** $f(x) = e^{5x+8}$	**c** $f(x) = e^{x+8}$
d $f(x) = 3e^{7x}$	**e** $f(x) = -6e^{9-\frac{x}{2}}$	**f** $f(x) = ke^{ax+b}$

Example Differentiate Exponential Functions of the form $f(x) = e^{ax+b}$

Differentiate:

$y = (1 + e^{2x})^3$

$y' = 3(1 + e^{2x}) \times 2e^{2x}$

$= 6e^{2x}(1 + e^{2x})$

$y = x^2e^{5x}$

$y = u'v + uv'$

$= 2xe^{5x} + x^2(5e^{5x})$

$= 2xe^{5x} + 5x^2e^{5x}$

$= xe^{5x}(2 + 5x)$

$y = \dfrac{x+1}{e^{3x}}$

$y' = \dfrac{u'v - uv'}{v^2}$

$= \dfrac{1(e^{3x}) - (x+1)(3e^{3x})}{(e^{3x})^2}$

$= \dfrac{e^{3x} - 3e^{3x}(x+1)}{e^{6x}}$

$= \dfrac{e^{3x}(1 - 3(x+1))}{e^{6x}}$

$= \dfrac{-2x - 3}{e^{3x}}$

Practice

Differentiate:

$y = x^3e^{-2x}$

$x^2e^{-2x}(3 - 2x)$

$y = (4 - e^{2x-7})^5$

$-10e^{2x-7}(4 - e^{2x-7})^4$

$y = \dfrac{e^x + 1}{3x + 2}$

$\dfrac{e^x(3x-1) - 3}{(3x+2)^2}$

1 Use the standard form $\frac{d}{dx}(e^{ax+b}) = ae^{ax+b}$ to differentiate:

a $y = e^{2x}$ $2e^{2x}$	**b** $y = e^{7x}$ $7e^{7x}$	**c** $y = e^{-x}$ $-e^{-x}$	**d** $y = -e^{5x}$ $-5e^{5x}$
e $y = e^{\frac{x}{2}}$ $\frac{1}{2}e^{\frac{x}{2}}$	**f** $y = 6e^{\frac{x}{3}}$ $2e^{\frac{x}{3}}$	**g** $y = e^{-\frac{x}{3}}$ $-\frac{1}{3}e^{-\frac{x}{3}}$	**h** $y = 5e^{\frac{x}{5}}$ $e^{\frac{x}{5}}$
i $f(x) = e^{x+2}$ e^{x+2}	**j** $f(x) = e^{x-3}$ e^{x-3}	**k** $f(x) = e^{5x+1}$ $5e^{5x+1}$	**l** $f(x) = e^{2x-1}$ $2e^{2x-1}$
m $f(x) = e^{-4x+1}$ $-4e^{-4x+1}$	**n** $f(x) = e^{-3x+4}$ $-3e^{-3x+4}$	**o** $f(x) = e^{-3x-6}$ $-3e^{-3x-6}$	**p** $f(x) = 2e^{\frac{x}{2}+4}$ $e^{\frac{x}{2}} + 4$

2 Differentiate:

a $f(x) = e^{x} + e^{-x}$ $e^{x} - e^{-x}$	**b** $f(x) = e^{2x} - e^{-3x}$ $2e^{2x} + 3e^{-3x}$	**c** $f(x) = \frac{e^{2x}}{2} + \frac{e^{3x}}{3}$ $e^{2x} + e^{3x}$
d $f(x) = \frac{e^{4x}}{4} + \frac{e^{5x}}{5}$ $e^{4x} + e^{5x}$	**e** $f(x) = \frac{e^{x} - e^{-x}}{2}$ $\frac{e^{x} + e^{-x}}{2}$	**f** $f(x) = \frac{e^{x} + e^{-x}}{3}$ $\frac{e^{x} - e^{-x}}{3}$

3 For each function, first find $\frac{dy}{dx}$ then evaluate $\frac{dy}{dx}$ at $x = 2$, giving your answer first in exact form and then correct to two decimal places.

a $y = e^{3x}$ $y' = 3e^{3x}$, $y'(2) = 3e^{6} = 1210.29$	**b** $y = e^{-2x}$ $y' = -2e^{-2x}$, $y'(2) = -2e^{-4} = -0.04$	**c** $y = e^{\frac{3x}{2}}$ $y' = \left(\frac{3}{2}\right)e^{\frac{3x}{2}}$, $y'(2) = \left(\frac{3}{2}\right)e^{3} = 30.13$

4 Differentiate

a $y = e^{5x} + e^{7x}$

$5e^{5x} + 7e^{7x}$

b $y = e^{4x+2} + e^{5+8x}$

$4e^{4x+2} + 8e^{5+8x}$

c $y = 4e^{-x} + 4e^{-3x}$

$-4e^{-x} - 12e^{-3x}$

d $y = 6e^{-2x-3} - 7e^{5-6x}$

$-12e^{-2x-3} + 42e^{5-6x}$

e $y = 5x^2 - 4x - 3e^{-x}$

$10x - 4 + 3e^{-x}$

f $y = e^{\frac{1}{2}x} + x^{\frac{1}{2}}$

$\frac{1}{2}e^{\frac{1}{2}x} + \frac{1}{2}x^{-\frac{1}{2}}$

5 Write each expression as a power of e, then differentiate.

a $y = \sqrt{e^x}$

$\frac{\sqrt{e^x}}{2}$

b $y = \sqrt[3]{e^x}$

$\frac{\sqrt[3]{e^x}}{3}$

c $y = \frac{1}{\sqrt{e^x}}$

$-\frac{1}{2\sqrt{e^x}}$

d $y = \frac{1}{\sqrt[3]{e^x}}$

$-\frac{1}{3\sqrt[3]{e^x}}$

6 Differentiate the following.

a $y = e^{\frac{2x-1}{3}}$

$\frac{2}{3}e^{\frac{2x-1}{3}}$

b $y = e^{\frac{2-3x}{5}}$

$-\frac{3}{5}e^{\frac{2-3x}{5}}$

c $y = (e^x)^3$

3e^(3x)

d $y = \frac{1}{e^x}$

$-\frac{1}{e^x}$

e $y = e^x\sqrt{e^x}$

$\frac{3}{2}e^{\frac{3x}{2}}$

f $y = \frac{e^x+1}{e^x}$

$-e^{-x}$

7 Differentiate:

a $y = 9e^x$

$9e^x$

b $y = -e^x$

$-e^x$

c $y = e^x + x^2$

$e^x + 2x$

d $y = 2x^3 - 3x^2 + 5x - e^x$

$6x^2 - 6x + 5 - e^x$

e $y = (e^x + 1)^3$

$3e^x(e^x + 1)^2$

f $y = (e^x + 5)^7$

$7e^x(e^x + 5)^6$

g $y = (2e^x - 3)^2$

$4e^x(2e^x - 3)$

h $y = xe^x$

$e^x(x + 1)$

i $y = \frac{e^x}{x}$

$\frac{e^x(x-1)}{x^2}$

j	$y = x^2e^x$	$xe^x(x + 2)$	**k**	$y = e^x(2x + 1)$	$e^x(2x + 3)$	**l**	$y = \frac{e^x}{7x - 3}$	$\frac{e^{x(7x-10)}}{(7x - 3)^2}$
m	$y = \frac{5x}{e^x}$	$\frac{5(1 - x)}{e^x}$	**n**	$y = e^{2x}$	$2e^{2x}$	**o**	$y = e^{-x}$	$-e^{-x}$
p	$y = 2e^{3x}$	$6e^{3x}$	**q**	$y = -e^{7x}$	$-7e^{7x}$	**r**	$y = -3e^{2x} + x^2$	$-6e^{2x} + 2x$
s	$y = e^{2x} - e^{-2x}$	$2(e^{2x} + e^{-2x})$	**t**	$y = 5e^{-x} - 3x + 2$	$-5e^{-x} - 3$	**u**	$y = xe^{4x}$	$e^{4x}(4x + 1)$
v	$y = \frac{2e^{3x} - 3}{x + 1}$	$\frac{6xe^{3x} + 4e^{3x} + 3}{(x + 1)^2}$	**w**	$y = (9e^{3x} + 2)^5$	$135e^{3x(9e^{3x}+2)^4}$			

8 Define the **hyperbolic** cosine and sine functions:

$$\cosh x = \frac{e^x + e^{-x}}{2} \text{ and } \sinh x = \frac{e^x - e^{-x}}{2}$$

a Show that $\frac{d}{dx}\cosh x = \sinh x$ and $\frac{d}{dx}\sinh x = \cosh x$.

$$\frac{d}{dx}\cosh x = \frac{e^x - e^{-x}}{2} = \sinh x$$
$$\frac{d}{dx}\sinh x = \frac{e^x - -e^{-x}}{2} = \cosh x$$

b Show that $\cosh^2 x - \sinh^2 x = 1$.

$$\cosh^2 x - \sinh^2 x = \left(\frac{e^x + e^{-x}}{2}\right)^2 - \left(\frac{e^x - e^{-x}}{2}\right)^2$$
$$= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4}$$
$$= \frac{e^{2x} + 2 + e^{-2x} - e^{2x} + 2 - e^{-2x}}{4}$$
$$= \frac{4}{4} = 1$$

Tangents and Normals of Exponential Functions

FOUNDATION

1

a Use calculus to find the gradient of the tangent to $y = e^x$ at Q(0, 1)

$\frac{dy}{dx} = e^x$

at $Q(0,1)$: $\frac{dy}{dx} = e^0 = 1$

b Hence find the equation of the tangent at Q.

$y - 1 = 1(x - 0)$

$y = x + 1$

c Find the x-intercept of the tangent.

At $y = 0$,

$x = -1$

2

a Use calculus to find the gradient of the tangent to $y = e^x$ at P(1, e)

$\frac{dy}{dx} = e^x$

at $P(1,e)$: $\frac{dy}{dx} = e^1 = e$

b Hence find the equation of the tangent at P.

$y - e = e(x - 1)$

$y = ex - e + e$

$y = ex$

c Find the x-intercept of the tangent.

At $y = 0$,

$x = 0$

3

a Use calculus to find the gradient of the tangent to $y = e^x$ at $R\left(-1, \frac{1}{e}\right)$

$\frac{dy}{dx} = e^x$

at $R\left(-1, \frac{1}{e}\right)$: $\frac{dy}{dx} = e^{-1} = \frac{1}{e}$

b Hence find the equation of the tangent at R.

$y - \frac{1}{e} = \left(\frac{1}{e}\right)(x - (-1))$

$y = \left(\frac{1}{e}\right)x + \frac{1}{e} + \frac{1}{e}$

$y = \left(\frac{1}{e}\right)x + \frac{2}{e}$

c Find the x-intercept of the tangent.

at $y = 0$,

$\frac{1}{e}x = -\frac{2}{e}$

$x = -2$

4

a Find the equation of the tangent to $y = e^x$ at $x = t$.

At $x = t$: point is (t, e^t),

gradient $= e^t$

Tangent: $y - e^t = e^t(x - t)$,

so $y = e^t(x - t + 1)$

b Hence show the x-intercept of this tangent is $t - 1$.

$y = 0$:

$0 = e^t(x - t + 1)$

$e^t \neq 0$, so $x - t + 1 = 0$,

$\therefore x = t - 1$

c Does this agree with what you found in Q1 – 3?

Yes, the x-intercept is always 1 less than the x-coordinate of the point of tangency.

5

a Explain why $y = e^x$ is always increasing.

$\frac{dy}{dx} = e^x > 0$

b Explain why $y = e^{-x}$ is always decreasing.

$\frac{dy}{dx} = -e^{-x} < 0$

6

a Differentiate $y = x - e^x$.

$\frac{dy}{dx} = 1 - e^x$

b Find the gradient of the tangent to $y = x - e^x$ at the point where $x = 1$.

Point of tangency at $x = 1$:

$y = 1 - e$, $\therefore (1, 1 - e)$

c Write down the equation of the tangent, and say why it passes through the origin.

Tangent: $y - (1 - e) = (1 - e)(x - 1)$

Expanding, simplifying, and refactorising:

$y = (1 - e)x$

This is in the form $y = kx$ so it must pass through the origin.

d Find the values of x for which the curve is:

i Stationary

$1 - e^x = 0$

$x = 0$

ii Increasing

$1 - e^x > 0$

$e^x < 1$

$x < 0$

iii Decreasing

$1 - e^x < 0$

$e^x > 1$

$x > 0$

7

a Write down the coordinates of the point R on the curve $y = e^{3x+1}$ where $x = -\frac{1}{3}$.

b Find $\frac{dy}{dx}$ and hence show that the gradient of the tangent at R is 3.

c What is the gradient of the normal at R?

d Hence find the equation of the normal at R in general form.

a) At $x = -\frac{1}{3}$: $y = e^{3\left(-\frac{1}{3}\right)+1} = 1$, $\therefore R\left(-\frac{1}{3}, 1\right)$

b) $\frac{dy}{dx} = 3e^{3x+1}$, at R: gradient $= 3e^0 = 3$

c) $-\frac{1}{3}$

d) Normal: $y - 1 = \left(-\frac{1}{3}\right)\left(x + \frac{1}{3}\right)$

$3y - 3 = -x - \frac{1}{3}$

$x + 3y = \frac{8}{3}$

8

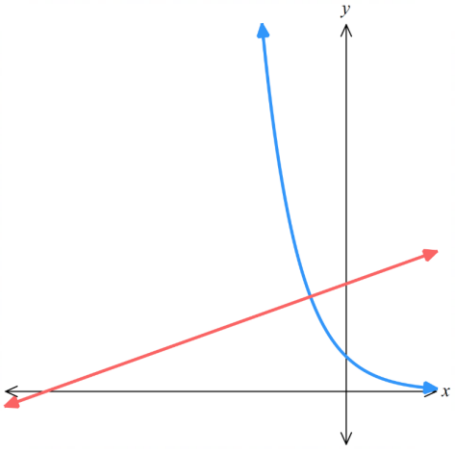
a Find the gradient of the tangent to $y = e^{-x}$ at the point $P(-1, e)$.

b Thus write down the gradient of the normal at this point.

c Hence determine the equation of this normal.

d Find the x- and y-intercepts of the normal.

e Find the area of the triangle whose vertices lie at the intercepts and the origin.



a) $\frac{dy}{dx} = -e^{-x}$, at $P(-1, e)$: gradient $= -e^1 = -e$

b) $\frac{1}{e}$

c) $y - e = \left(\frac{1}{e}\right)(x + 1)$,

so $y = \left(\frac{1}{e}\right)x + \frac{1}{e} + e$

d) x intercept: $0 = \left(\frac{1}{e}\right)x + \frac{1}{e} + e$, so $x = -1 - e^2$

y intercept: $y = \frac{1}{e} + e$

e) Area $= \left(\frac{1}{2}\right) \times |1 + e^2| \times \left|\frac{1}{e} + e\right| = \left(\frac{1}{2}\right)(1 + e^2)\frac{(1 + e^2)}{e} = \frac{(1 + e^2)^2}{2e}$

9

a Use the derivative to find the gradient of the tangent to $y = e^x$ at $B(0,1)$.

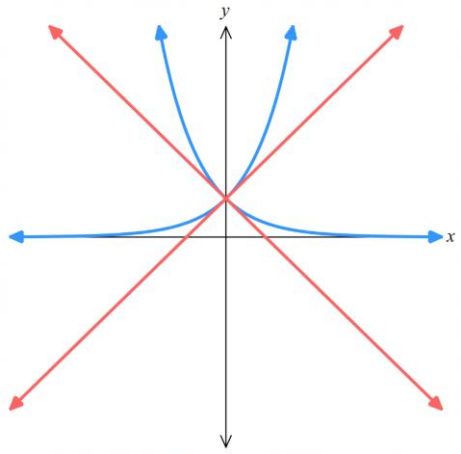
b Hence find the equation of this tangent and show that it meets the x-axis at $F(-1,0)$.

c Use the derivative to find the gradient of the tangent to $y = e^{-x}$ at $B(0,1)$.

d Hence find the equation of this tangent and show that it meets the x-axis at $G(1,0)$.

e Sketch $y = e^x$ and $y = e^{-x}$ on the same set of axes, showing the two tangents.

f What sort of triangle is $\triangle BFG$, and what is its area?



a) $\frac{dy}{dx} = e^x$, at $B(0,1)$: gradient $= 1$

b) $y - 1 = 1(x - 0)$,

$y = x + 1$

At x axis: $0 = x + 1, so\ x = -1, \therefore F(-1,0)$

c) $\frac{dy}{dx} = -e^{-x}$, at $B(0,1)$: gradient $= -1$

d) $y - 1 = -1(x - 0)$

$y = -x + 1$

At x axis: $0 = -x + 1,$

$x = 1, \therefore G(1,0)$

f) $\triangle BFG$ is isosceles right triangle with area $= \left(\frac{1}{2}\right) \times 2 \times 1 = 1$

10

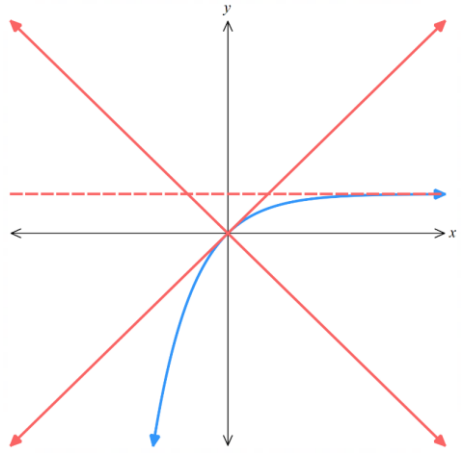
a Show that the equation of the tangent to $y = 1 - e^{-x}$ at the origin is $y = x$.

b Deduce the equation of the normal at the origin without further use of calculus.

c What is the equation of the asymptote of this curve?

d Sketch the curve, showing the points T and N where the tangent and normal respectively cut the asymptote.

e Find the area of $\triangle OTN$.



a) At $(0,0)$: $\frac{dy}{dx} = e^{-x}$, gradient $= e^0 = 1$

Tangent: $y - 0 = 1(x - 0)$

$y = x$

b) Normal gradient $= -1$, so normal: $y = -x$

c) Asymptote: $y = 1$

d) T where $y = x$ meets $y = 1$:

$x = 1, so\ T(1,1)$

N where $y = -x$ meets $y = 1$:

$-x = 1, so\ N(-1,1)$

e) Area of $\triangle OTN = \left(\frac{1}{2}\right) \times 2 \times 1 = 1$

Chain Rule with Exponential Functions

Review

☐ Differentiate exponential functions base e.

a $f(x) = e^x$	**b** $f(x) = e^{2x}$	**c** $f(x) = e^{3x-4}$
d $f(x) = e^{11-7x}$	**e** $f(x) = e^{(\ln 5)x}$	**f** $f(x) = e^7$

Investigation Chain rule for exponential functions.

Consider $y = e^{f(x)}$.

Using chain rule, find the general form of the derivative.

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Let $u = f(x)$

$$\frac{du}{dx} =$$

Let $y = e^u$

$$\frac{dy}{du} =$$

ⓘ Chain Rule for Exponential Functions

$$y = e^{f(x)}$$

$$\frac{dy}{dx} = f'(x)\, e^{f(x)}$$

Identify the derivative of an exponential function.

$f(x) = e^{3x+5}$ $f'(x) = 3e^{3x+5}$	$f(x) = e^{x^2}$ $f'(x) = 2x\, e^{x^2}$	$f(x) = e^{7x^3+5x^2+3}$ $f'(x) = (21x^2 + 10x)\, e^{7x^3+5x^2+3}$

Differentiate these functions.

a $f(x) = e^{2x+5}$	**b** $f(x) = e^{-4x^2}$	**c** $f(x) = e^{10x^2-11x}$
d $f(x) = e^{x^2-2}$	**e** $f(x) = e^{15x^3-3x}$	**f** $f(x) = e^{\frac{1}{x^2}}$

Example Differentiate exponential functions base e.

Differentiate

$y = (e^{2x} - 3)^4$ use chain rule

$y' = 4(e^{2x} - 3)^3 \times (2e^{2x})$

$= 8e^{2x}(e^{2x} - 3)^3$

Differentiate

$y = x^3 e^{1-x^2}$ use product rule

$y' = u'v + uv'$

$= 3x^2(e^{1-x^2}) + x^3(-2xe^{1-x^2})$

$= 3x^2(e^{1-x^2}) - 2x^4(e^{1-x^2})$

$= x^2(e^{1-x^2})(3 - 2x^2)$

Practice

Differentiate

$y = x^3 e^{5x^2}$

$x^2 e^{5x^2}(10x^2 + 3)$

Differentiate

$y = (e^x + 2x)^3$

$3(e^x + 2)(e^x + 2x)^2$

Differentiate

$y = \dfrac{e^{2x+3}}{1 - x}$

$\dfrac{e^{2x+3}(3 - 2x)}{(1 - x)^2}$

Differentiate

$y = e^{3-x^2}(1 - x)$

$e^{3-x^2}(2x^2 - 2x - 1)$

1 Use the chain rule to differentiate:

a $y = e^{ax+b}$

$\frac{dy}{dx} = ae^{ax+b}$

b $y = e^{x^2}$

$\frac{dy}{dx} = 2xe^{x^2}$

c $y = e^{-\frac{x^2}{2}}$

$\frac{dy}{dx} = -xe^{-\left(\frac{1}{2}\right)x^2}$

d $y = e^{x^2+1}$

$\frac{dy}{dx} = 2xe^{x^2+1}$

e $y = e^{1-x^2}$

$\frac{dy}{dx} = -2xe^{1-x^2}$

f $y = e^{x^2+2x}$

$\frac{dy}{dx} = 2(x + 1)e^{x^2+2x}$

g $y = e^{6+x-x^2}$

$\frac{dy}{dx} = (1 - 2x)e^{6+x-x^2}$

h $y = \frac{e^{3x^2-2x+1}}{2}$

$\frac{dy}{dx} = \left(\frac{1}{2}\right)(6x - 2)e^{3x^2-2x+1}$
$= (3x - 1)e^{3x^2-2x+1}$

2 Use the product rule to differentiate:

a $y = xe^x$

$y' = e^x + xe^x = (x + 1)e^x$

b $y = xe^{-x}$

$y' = e^{-x} - xe^{-x}$
$= (1 - x)e^{-x}$

c $y = (x - 1)e^x$

$y' = e^x + (x - 1)e^x$
$= xe^x$

d $y = (x + 1)e^{3x-4}$

$y' = e^{3x-4} + 3(x + 1)e^{3x-4}$
$= (3x + 4)e^{3x-4}$

e $y = x^2e^{-x}$

$y' = 2xe^{-x} - x^2e^{-x}$
$= (2x - x^2)e^{-x}$

f $y = (2x - 1)e^{2x}$

$y' = 2e^{2x} + 2(2x - 1)e^{2x}$
$= 4xe^{2x}$

g $y = (x^2 - 5)e^x$

$y' = 2xe^x + (x^2 - 5)e^x$
$= (x^2 + 2x - 5)e^x$

h $y = x^3e^{2x}$

$y' = 3x^2e^{2x} + 2x^3e^{2x}$
$= x^2e^{2x}(3 + 2x)$

3 Use the quotient rule to differentiate:

a $y = \frac{e^x}{x}$

$y' = \frac{xe^x - e^x}{x^2}$
$= e^x\left(\frac{x-1}{x^2}\right)$

b $y = \frac{x}{e^x}$

$y' = \frac{e^x - xe^x}{e^{2x}}$
$= (1-x)e^{-x}$

c $y = \frac{e^x}{x^2}$

$y' = \frac{x^2e^x - 2xe^x}{x^4}$
$= \left(\frac{x-2}{x^3}\right)e^x$

d $y = \frac{x^2}{e^x}$

$y' = \frac{2xe^x - x^2e^x}{e^{2x}}$
$= (2x - x^2)e^{-x}$

e $y = \frac{e^x}{x+1}$

$y' = \frac{(x+1)e^x - e^x}{(x+1)^2}$
$= \frac{xe^x}{(x+1)^2}$

f $y = \frac{x+1}{e^x}$

$y' = \frac{e^x - (x+1)e^x}{e^{2x}}$
$= -xe^{-x}$

g $y = \frac{x-3}{e^{2x}}$

$y' = \frac{e^{2x} - 2(x-3)e^{2x}}{e^{4x}}$
$= (7 - 2x)e^{-2x}$

h $y = \frac{1-x^2}{e^x}$

$y' = \frac{-2xe^x - (1-x^2)e^x}{e^{2x}}$
$= (x^2 - 2x - 1)e^{-x}$

4 Expand and simplify each expression, then differentiate it.

a $(e^x + 1)(e^x + 2)$

$y' = 2e^{2x} + 3e^x$

b $(e^{2x} + 3)(e^{2x} - 2)$

$y' = 4e^{4x} + 2e^{2x}$

c $(e^{-x} + 2)(e^{-x} + 4)$

$y' = -2e^{-2x} - 6e^{-x}$

d $(e^{-3x} - 1)(e^{-3x} - 5)$

$y' = -6e^{-6x} + 18e^{-3x}$

e $(e^{2x} + 1)(e^x + 1)$

$y' = 3e^{3x} + 2e^{2x} + e^x$

f $(e^{3x} - 1)(e^{-x} + 4)$

$y' = 2e^{2x} + 12e^{3x} + e^{-x}$
$= 12e^{3x} + 2e^{2x} + e^{-x}$

5 Use the chain rule to differentiate:

a $y = (1 - e^x)^5$

$y' = -5e^x(1 - e^x)^4$

b $y = (e^{4x} - 9)^4$

$y' = 16e^{4x}(e^{4x} - 9)^3$

c $y = \frac{1}{e^x - 1}$

$y' = -\frac{e^x}{(e^x - 1)^2}$

d $y = \frac{1}{(e^{3x} + 4)^2}$

$y' = -\frac{6e^{3x}}{(e^{3x} + 4)^3}$

6

a Show by substitution that the function $y = 3e^{2x}$ satisfies the equation $\frac{dy}{dx} = 2y$.

$y = 3e^{2x}$,

$\frac{dy}{dx} = 6e^{2x} = 2(3e^{2x}) = 2y$

b Show by substitution that the function $y = 5e^{-4x}$ satisfies the equation $\frac{dy}{dx} = -4y$.

$y = 5e^{-4x}$,

$\frac{dy}{dx} = -20e^{-4x} = -4(5e^{-4x}) = -4y$

7 For the curve $y = e^{x^2}$ at $(1, e)$, find the equation of the tangent and normal in general form.

Tangent: $2ex - y - e = 0$

Normal: $x + 2ey - 2e^2 - 1 = 0$

8 Use the product, quotient and chain rules as appropriate to differentiate:

a $y = (e^x + 1)^3$

$y' = 3e^x(e^x + 1)^2$

b $y = (e^x + e^{-x})^4$

$y' = 4(e^x - e^{-x})(e^x + e^{-x})^3$

c $y = (1 + x^2)e^{1+x}$

$y' = 2(1 + x)e^{1+x} + (1 + x)^2e^{1+x}$
$= (1 + x)^2e^{1+x}$

d $y = (x^2 - x)e^{2x-1}$

$y' = 4xe^{2x-1} + 2(2x^2 - 1)e^{2x-1}$
$= (4x^2 + 4x - 2)e^{2x-1}$
$= 2(2x^2 + 2x - 1)e^{2x-1}$
$= (2x^2 - 1)e^{2x-1}$

e $y = \frac{e^x}{e^x + 1}$

$y' = \frac{e^{x(e^x+1)} - e^{x(e^x)}}{(e^x + 1)^2}$
$= \frac{e^x}{(e^x + 1)^2}$

f $y = \frac{e^x + 1}{e^x - 1}$

$y' = \frac{e^{x(e^x-1)} - e^{x(e^x+1)}}{(e^x - 1)^2}$
$= -\frac{2e^x}{(e^x - 1)^2}$

9 Write each expression as the sum of simple powers of e, then differentiate it.

a $y = \frac{e^x + 1}{e^x}$

$y = 1 + e^{-x}$
$y' = -e^{-x}$

b $y = \frac{e^{2x} + e^x}{e^x}$

$y = e^x + 1$
$y' = e^x$

c $y = \frac{2 - e^x}{e^{2x}}$

$y = 2e^{-2x} - e^{-x}$
$y' = -4e^{-2x} + e^{-x}$
$= e^{-x} - 4e^{-2x}$

d $y = \frac{3 + e^x}{e^{4x}}$

$y = 3e^{-4x} + e^{-3x}$
$y' = -12e^{-4x} - 3e^{-3x}$

e $y = \frac{e^x + e^{2x} - 3e^{4x}}{e^x}$

$y = 1 + e^x - 3e^{3x}$
$y' = e^x - 9e^{3x}$

f $y = \frac{e^{2x} + 2e^x + 1}{e^{2x}}$

$y = 1 + 2e^{-x} + e^{-2x}$
$y' = -2e^{-x} - 2e^{-2x}$

10 Differentiate these functions.

a $y = \sqrt{e^x}$

$y = e^{\frac{x}{2}}$

$y' = \frac{e^{\frac{x}{2}}}{2}$

$= \frac{\sqrt{e^x}}{2}$

b $y = \sqrt[3]{e^x}$

$y = e^{\frac{x}{3}}$

$y' = \frac{e^{\frac{x}{3}}}{3}$

$= \frac{\sqrt[3]{e^x}}{3}$

c $y = \frac{1}{\sqrt{e^x}}$

$y = e^{-\frac{x}{2}}$

$y' = -\frac{e^{-\frac{x}{2}}}{2}$

$= -\frac{1}{2\sqrt{e^x}}$

d $y = \frac{1}{\sqrt[3]{e^x}}$

$y = e^{-\frac{x}{3}}$

$y' = -\frac{e^{-\frac{x}{3}}}{3}$

$= -\frac{1}{3\sqrt[3]{e^x}}$

e $e^{\sqrt{x}}$

$y' = \frac{e^{\sqrt{x}}}{2\sqrt{x}}$

f $e^{-\sqrt{x}}$

$y' = -\frac{e^{-\sqrt{x}}}{2\sqrt{x}}$

g $e^{\frac{1}{x}}$

$y' = -\frac{e^{\frac{1}{x}}}{x^2}$

h $e^{-\frac{1}{x}}$

$y' = \frac{e^{-\frac{1}{x}}}{x^2}$

11 An object moves according to $x = \frac{e^{t-0.1t^3}}{2}$

a find its displacement and velocity $t = 2$ seconds.

Displacement at t = 2:

x(2) ≈ 1.66 m

Velocity:

$\frac{dx}{dt} = \frac{(1 - 0.3t^2)(e^{t-0.1t^3})}{2}$

$v(2) \approx -0.332$ m/s

b what is its displacement and velocity as $t \to \infty$?

Displacement and velocity both approach 0.

Derivative of Exponential Functions of Any Base

Review

☐ Differentiate exponential functions base e.

a $f(x) = e^{(\ln 5)x}$	**b** $f(x) = e^{(\log_3 7)x}$	**c** $f(x) = e^{(\log_5 13^x)}$

☐ Change the base of an exponential using $e^{\log_e a} = a$.

a $e^{\log_e 7} =$	**b** $e^{\ln 5} =$	**c** $12 = e^{(\quad)}$
d $12^x = e^{(\quad)}$	**e** $7^x =$	**f** 7^{-x}

Investigation Derivative of exponential functions of any base.

We know that $\frac{d(e^{ax+b})}{dx} = ae^{ax+b}$; our rule currently only works for base e.

Consider $f(x) = 2^x$, to find the derivative of this function, we can express 2^x with base e by using the identity $e^{\log_e a} = a$:

$$f(x) = 2^x = e^{(\log_e 2^x)}$$
$$= e^{x \log_e 2}$$

$$f'(x) = (\log_e 2)e^{x \log_e 2}$$
$$= (\ln 2) \cdot 2^x$$

Follow the steps above to differentiate:

$f(x) = 10^x$

$f'(x) =$

ⓘ Derivative of Exponential Functions of Any Base

Every positive real number can be written with base e: $a = e^{\ln a}$

Every exponential function can be written with base e: $a^x = e^{x \ln a}$

To differentiate exponential functions of any base:

$$f(x) = a^x$$
$$f'(x) = (\ln a)\, a^x$$

Identify the derivative of an exponential function.

$f(x) = 11^x$ $f'(x) = (\ln 11)\, 11^x$	$f(x) = 13^x$ $f'(x) = (\ln 13)\, 13^x$	$f(x) = 3^x$ $f'(x) = (\ln 3)\, 3^x$

Differentiate these functions.

a $f(x) = 17^x$	**b** $f(x) = 25^x$	**c** $f(x) = 716^x$
d $f(x) = 5 \times 5^x$	**e** $f(x) = 5^{x+1}$	**f** $f(x) = (-3)^x$

ⓘ Investigation Chain rule for exponential functions of any base.

Consider $y = a^{f(x)}$.

Using chain rule, find the general form of the derivative.

$y = a^{f(x)}$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Let $u = f(x)$

$$\frac{du}{dx} =$$

Let $y = a^u$

$$\frac{dy}{du} =$$

ⓘ Chain Rule for Exponential Functions of Any Base

$$y = a^{f(x)}$$

$$\frac{dy}{dx} = f'(x)\,(\ln a)\,a^{f(x)}$$

Identify the derivative of an exponential function.

$f(x) = 11^{13x+5}$ $f'(x) = 13(\ln 11)11^{13x+5}$	$f(x) = 13^{x^2}$ $f'(x) = 2x\,(\ln 13)\,13^{x^x}$	$f(x) = 5^{x^3-7x+3}$ $f'(x) = (3x^2-7)(\ln 5)5^{x^3-7x+3}$

Differentiate these functions.

a $f(x) = 3^{2x+5}$	**b** $f(x) = 7^{-4x}$	**c** $f(x) = 8^{x-11}$
d $f(x) = 7^{x^2}$	**e** $f(x) = 12^{5x^2+11}$	**f** $f(x) = 19^{3x^5-12x}$

1 Express these functions as powers of e, then differentiate them.

a $y = 3^x$

$y = 3^x = e^{x \ln 3}$

$\frac{dy}{dx} = 3^x \ln 3$

b $y = 4^x$

$y = 4^x = e^{x \ln 4}$

$\frac{dy}{dx} = 4^x \ln 4$

c $y = 2^x$

$y = 2^x = e^{x \ln 2}$

$\frac{dy}{dx} = 2^x \ln 2$

2 Differentiate using the formula.

a $y = 10^x$

$\frac{dy}{dx} = 10^x \ln 10$

b $y = 8^x$

$\frac{dy}{dx} = 8^x \ln 8$

c $y = 3 \times 5^x$

$\frac{dy}{dx} = 3(\ln 5)(5^x)$

3 Differentiate $y = 2^{3x-4}$.

$\frac{dy}{dx} = 3(2^{3x-4}) \ln 2$

4 Find the equation of the tangent to the curve $y = 4^{x+1}$ when $x = 0$. Answer in general form.

At $x = 0, y = 4$

$\frac{dy}{dx} = 4^{x+1} \ln 4$

$At\ (0,4): m = 4\ ln\ 4$

$y - y_1 = m(x - x_1)$:

$y - 4 = 4 \ln 4(x - 0)$

$y = 4x \ln 4 + 4$

$4x \ln 4 - y + 4 = 0$